Emmanuel Acheampong Safo

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ABOUT ME

I am a Master's student in Electrical Engineering at Tampere University, specializing in Signal Processing and Machine Learning. My technical foundation is built on completed coursework in Computational Imaging, 3D and Extended Reality, and Audio Processing. Currently, I am advancing my expertise in Computer Vision and Advanced Deep Learning while developing an Intelligent Cry-Detection System.
Beyond academia, I bring practical experience architecting scalable software. I developed High Purchase, a multi-tenant FinTech platform featuring real-time analytics for over 500 businesses. Proficient in Python, C++, and Next.js, I seamlessly bridge complex algorithms with functional, real-world applications.

EDUCATION AND TRAINING

18/08/2025 – CURRENT Tampere, Finland
MASTERS IN COMPUTER SCIENCE AND ELECTRICAL ENGINEERING SPECIALIZING IN SIGNAL PROCESSING AND MACHINE LEARNING Tampere University

Website https://www.tuni.fi/en | **Level in EQF** EQF level 7

21/08/2018 – 16/06/2022 Accra, Ghana
BSC ELECTRICAL/ELECTRONICS ENGINEERING Regional Maritime University

Website https://rmu.edu.gh/ | **Level in EQF** EQF level 6

09/06/2021 – 18/06/2022 Ghana
ONLINE COURSE IN SOFTWARE DEVELOPMENT SPECIALIST IN MACHINE LEARNING, WEB APPLICATION WITH NODEJS AND NEXTJS Udemy

Website https://www.udemy.com/

WORK EXPERIENCE

SENIOR SOFTWARE DEVELOPER – SYAAK.INC – 09/08/2025 – Current – ACCRA, GHANA

- Architected Multi-Tenant FinTech Solutions: Developed the High Purchase BNPL (Buy Now, Pay Later) platform, enabling 500+ Ghanaian businesses to manage multi-shop operations, inventory, and automated hire-purchase agreements from a unified dashboard.
- Engineered Real-Time Payment Systems: Integrated secure payment collection modules and automated SMS/Email invoicing gateways, contributing to a 75% reduction in manual tracking errors and lowering customer default rates.
- Developed Data-Driven Credit Scoring: Built customer tracking and behavior analysis modules to generate credit scores, helping merchants assess lending risks using historical payment data.
- Optimized Mobile-First Web Applications: Leveraged Next.js and modern web technologies to build high-performance, "no-app-needed" mobile experiences tailored for low-bandwidth environments in the Ghanaian market.
- Built Scalable Analytics Dashboards: Designed and implemented real-time financial reporting engines that translate raw transaction data into actionable business intelligence (BI) for shop owners.
- Maintained High-Availability Infrastructure: Ensured 99.9% system uptime for mission-critical operations, including real-time debt collection tracking and instant payment confirmations.
- Bridged EdTech and Data Science: Supported the development of the SYAAK learning platform, facilitating hands-on training for 500+ students in Python, R, and Machine Learning.

Currently Project: https://high-purchase.vercel.app/
https://syaak.vercel.app/

ELECTRO TECHNICAL OFFICER – DANAOS SHIPPING CO. LTD – 01/01/2024 – 11/05/2025 – PIREAS, GREECE

Electrical Maintenance & Repairs
- Supervised and maintained electrical power distribution, switchboards, and high-voltage systems.
- Diagnosed and repaired faults in motors, generators, transformers, and batteries.
- Conducted routine inspections and preventive maintenance to ensure system reliability.

Automation & Control Systems
- Monitored and troubleshot automated systems, including ballast control and HVAC.
- Ensured seamless operation of engine control and monitoring systems.

Navigation & Communication Equipment
- Maintained and repaired radar, GPS, ECDIS, AIS, and other critical navigation systems.
- Ensured smooth operation of shipboard communication systems (VHF, MF/HF, satellite).

Energy Efficiency & Power Management
- Monitored vessel power consumption and optimized electrical efficiency.
- Implemented fuel-saving initiatives related to electrical operations.

Compliance & Documentation
- Ensured compliance with maritime safety regulations (SOLAS, ISM, ISPS).
- Maintained records of electrical maintenance, repairs, and safety drills.

UNIVERSITY TEACHING ASSISTANT – REGIONAL MARITIME UNIVERSITY – 06/06/2022 – 14/06/2023 – ACCRA, GHANA

- Supervised Laboratory Experiments: Instructed students on the proper setup and operation of essential telecommunication hardware, including spectrum analyzers, digital oscilloscopes, and function generators, ensuring safety and accuracy.
- Facilitated Software Simulations: Guided students through designing and testing communication systems (such as AM/FM/PSK modulation) using industry-standard simulation tools like MATLAB/Simulink or LabVIEW.
- Clarified Theoretical Concepts: Conducted tutorial sessions to break down complex lecture topics, specifically focusing on Fourier analysis, signal processing, and electromagnetic theory to improve student comprehension.
- Provided Technical Troubleshooting: Assisted students during practical sessions by debugging faulty circuit connections and analyzing signal integrity issues to ensure successful experiment completion.
- Mentored Student Projects: Offered guidance on capstone or term projects, helping students apply theoretical knowledge to practical design challenges in wireless or digital communications

LANGUAGE SKILLS

Mother tongue(s): **ENGLISH**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
FINNISH	B1	B1	A2	A2	B1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

SKILLS

Scientific Computing & Simulation

C/C++ | MATLAB & Simulink | CMake

Embedded Systems & Hardware Description

Verilog: | Atmel Assembly & C | C++ | Internet of Things

Industrial Automation & Smart Grid Technologies

Power Management Systems (PMS) | Hardware Interface | SCADA Automation

Machine Learning, Deep Learning, Computer Vision & Data Visualization

Tensorflow(Python) | Python for Machine Learning: Pytorch | Supervised and Unsupervised Learning | OpenCV | Yolo | Python (Pandas)

Software Development

Nextjs | Typescript | Git | Database Management: MySQL/PostgreSQL | Nodejs

Professional & Interpersonal Skills

Cross-Functional Collaboration | Crisis Management & Adaptability | Technical Mentorship | Intercultural Communication | Safety Practices

Database

SQL | PostGreSQL | MongoDB

RECOMMENDATIONS

Ing. Prophet Dr. Freddy Kwami Ayatey Senior Lecture

- I had the privilege of teaching him from level 100 to level 400
- He is wonderful in team work, learning quick and one of the most intelligent students that i have taught throughout my teaching carrer
- I strongly recommend for any role or scholarship

Email ayateyfreddy@gmail.com | **Phone** (+233) 243848052

Ing. Emmanuel Yao Wormenor Senior Engineer/Lecturer

- I have know them for several years both in classroom and professionally
- He is an excellent thinker, good programmer and an excellent team leader
- I strongly recommend Safo Emmanuel

Email wey81@yahoo.co.uk | **Phone** (+233) 243119529

PROJECTS

09/01/2022 – 04/04/2022

A MACHINE LEARNING APPROACH TO ELECTRICAL DEVICE SIGNATURE IDENTIFICATION

- Effective energy management in residential buildings, particularly school hostels, is vital for cost control.
- Developed system incorporates a multi-socket, smart meter, relay, and algorithms for real-time energy monitoring and control.
- Data from devices is collected by the energy meter and processed on a laptop using a K-NN algorithm.
- System identifies and shuts down unwanted devices while maintaining essential ones.
- Goals include cost reduction, creation of consumption profiles, and efficient energy management.
- System exhibited successful performance during testing, showcasing its effectiveness.

10/01/2021 – 14/04/2021

DESIGN AND CONSTRUCTION OF A LOW-SCALE AUTOMATIC GENERATOR MANAGEMENT SYSTEM

- Automated system for managing small-scale generators, optimizing their operation and maintenance through intelligent design and construction.
- Incorporates sensors and software to monitor generator status, control power output, and schedule maintenance, enhancing efficiency and reliability.
- Tailored for small-scale applications, offering seamless management of power generation, minimizing downtime and ensuring consistent performance.

HONOURS AND AWARDS

06/06/2022

Best Final year Electrical Department Project – Regional Maritime University

- A cash prize award for the most impactful final year project